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Welfare evaluation of subsidies to renewable energy in general equilibrium: Theory and application☆,☆☆

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ABSTRACT

While support schemes to renewable energy are ubiquitous around the world today, there are few systematic welfare evaluations of their social benefits and costs in an economy-wide setting. We develop a general equilibrium cost–benefit rule to assess changes in quantity based subsidy schemes, "green" certificates, that support renewable electricity generation. An advantage to large-scale numerical models of the same issue is that we can go "into the black box" and uncover key economic mechanisms. We study a second-best economy with distorting taxes and pollution, so that a perturbation of the certificate scheme causes both benefits and costs; these items can be uncovered and estimated using our framework. To this end, we provide a user-friendly approximation for empirical implementation, which means that data requirement is modest relative to a typical computable general equilibrium model. We apply the theory to a currently existing scheme in Sweden taking into account "trickle-down" effects, including e.g. a loss of value-added tax income in the rest of the economy and environmental costs (i.e. externalities from electricity generation not currently internalized). We first present an ex post estimate, i.e. the welfare consequences of having scrapped the existing system 2003–2017 and then an ex ante analysis of extending the system to 2045. The latter includes a systematic sensitivity analysis based on Monte-Carlo simulation. Overall, we find net present value gains from removing the subsidy scheme, taking into account externalities, "trickle-down" and public finance repercussions.

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1. Introduction

According to Hirth (2017), various types of "green electricity" subsidy schemes exist in 145 countries, including India's *National Solar Plan*, the German *Energiwende* and USA's *Clean Power Plan*. The dominating forms of support are feed-in tariffs and "green" certificates, i.e. price or quantity supports.[1] We focus certificates that (like feed-in tariffs) have been introduced to promote electricity generation based on renewable resources, such as wind power and solar panels.[2] Through a market mechanism, they guarantee that a stipulated proportion of generated electricity come from certified renewable sources.[3]

To the best of our knowledge, there are few, if any, cost–benefit analyses of support to "green" electricity that derive and estimate

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[1] Other popular forms of support include production tax credits and investment tax credits.

[2] Such certificates are also known as renewable portfolio standards (RPS, e.g., in the US), renewable energy certificates/credits or renewable obligation certificates (e.g., in the UK).

[3] Because there is typically only one (market) price of a certificate, this type of subsidy is technology neutral. By contrast, feed-in tariffs can easily be tailored to any supported technology.

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general equilibrium welfare measures.[4] Several numerical simulation models exist[5], but few, if any, papers present an integrated approach based on general equilibrium welfare theory.

A substantial literature (which we briefly review below) exists on the economics of replacing fossil fueled generation with renewable sources. These papers are typically based on partial equilibrium models and have significant detail on generation technology and externality assessment. This paper has a different focus. We consider three aspects that have received less attention in welfare evaluations of the current efforts to transform energy systems. First, we derive a cost–benefit rule based on general equilibrium theory to obtain some insights into the main mechanisms involved. It highlights, for example, public finance aspects and an indirect effect on aggregate demand.[6] Second, we provide a user friendly approximation of a complex equilibrium economy that makes the empirical implementation of our model straightforward. Third, our evaluation includes both public finance aspects *and* externalities.

In the next section (Section 2), we discuss the benefits and costs of subsidies to electricity generation, by scrutinizing three salient arguments (externalities, "jobs" and energy security) put forward in the debate. We then turn in Section 3 to a review of some of the empirical studies that exists on our topic of interest. Section 4 presents a social welfare function used to derive a general equilibrium cost–benefit rule for changes in green certificate schemes. Our model has electricity, an aggregate composite commodity, both subject to at least a value added tax (VAT), and public bads (harmful emissions). No further disaggregation is needed, because small price changes "net out" from the welfare measure via the envelope theorem. In this section, we also derive an approximation of the general result for convenient use in empirical work. Section 5 describes our application, which has two parts. We begin by focussing on the electricity market and provide three sets of results on consumer, producer surplus and public finance repercussions in this market. Thus, an ex post calculation displays the benefits and costs of having had the system in operation during 2003–2017, while an ex ante assessment presents a cost–benefit analysis of scrapping the system in the years 2018–2045. In the second part, we add economy-wide public finance aspects and externalities; a phasing out of the certificate scheme causes the demand curve for the composite commodity to shift to the left and raises the demand for replacement power (which, in our case, is most likely to be based on fossil fuels). Section 5.5 has a systematic sensitivity analysis in which we assess our assumptions about future prices and externality values by a Monte-Carlo simulation. We end by some remarks about how the analysis can be extended. Appendix A has additional details on the theory and the empirical analysis.

2. Why subsidize "green" electricity generation?

Several arguments have been advanced in favor of supporting subsidies to "green" electricity generation, see e.g. Schmalensee (2012), Borenstein (2012) and Fischer and Preonas (2010) for extensive reviews and critical analysis. We focus three main arguments that has been discussed in the literature: externalities, jobs and energy security. We begin, however, with arguments that discuss the use of subsidies in environmental policy when there already exists a first-best instrument, because this is relevant to understanding the welfare assessment in our application.

That carbon prices are "too low" in the EU-ETS[7] is a recurring theme in a special issue (Hepburn et al. 2016) on the so-called market stability reserve (MSR) (a mechanism that adjusts the supply of permits). In summing up the EU-ETS issues in general and the MSR in particular, the editors of the special issue comments:

> … it [EU-ETS] is sometimes described as having 'failed' because prices are too low to incentivise substantial short-run emissions reductions and too volatile to provide adequate long-run incentives for investments in clean technologies (p. 1)…has the MSR given the EU ETS the shot in the arm as intended? As of 2016, it might seem not. EUA prices remain low. Market confidence appears limited (p. 5).

Under a EU-agreement (9 Nov 2017), several measures are to be put in place which will play up the importance of the MSR for the permit price. Beyond cutting the supply of permits the agreement includes a mechanism for automatic cancellation. It is expected that these measures will have a significant impact on the carbon permit prices in the next decade. The automatic cancellation mechanism implies a window of time in which individual country measures to reduce their carbon emissions will have an effect on the cap. According to KI (2018), there is a window of opportunity for Sweden up until 2023, when domestic measures within the EU-ETS can affect the total emissions (at a maximum rate of 80%, the rate decreases over time to about 65% to 2023). Buying permits and scrapping them remains the most cost-effective option. We do not take the changes of EU-ETS into account in our analysis, assuming that the effect on the key prices by Swedish measures will be small.

While carbon pricing, according to basic theory, stimulate renewables, many countries in the EU have added various stimulus packages to speed up the introduction of renewable energy. The combined effect of such overlapping regulations is, however, not clear. Böhringer and Rosendahl (2010) argue that a combination of "black" and "green quotas", (e.g. EU-ETS and certificates) rather "promote black production" compared to a case when only "black quotas" are used. Intuitively, there will be a downward pressure on carbon permit prices due to the "green" subsidies, and this will be most beneficial for the "dirtiest" fossil fueled power plants. On the other hand, the subsidies might indirectly have an effect on the cap itself, as supported technologies replace "dirtier" ones. It is hard to evaluate such a claim, not the least without specifying what the alternative is (e.g. a cost–benefit analysis of tightening the cap).

2.1. Externalities

A standard economic argument for subsidizing an economic activity is based on the idea of positive externalities. In this context, the arguments essentially bifurcate into two strains, one relating to innovations, the other to environmental gains. The so-called Porter-argument (Porter, 1991) has a different starting point; in essence, stricter (environmental) regulation triggers innovation that otherwise would not have been made, which may turn into a first-mover advantage in export markets. A more stringent environmental policy could thus work as an indirect support to emission-free technologies. The empirical evidence supporting the Porter-argument is, at best, ambiguous (Böhringer et al., 2014).

Turning to direct support, if an innovator cannot capture the social return of her investment, she confers a positive externality on society. The classic case involves supporting "infant industries" in their early stages since it may reap rewards through innovations.

[4] Cavicchi (2017, p. 2) notes that for the US: " Very little social welfare cost–benefit analysis lies behind federal and state RPS programs designed to subsidize renewable electricity resources."

[5] See e.g. Abrell et al. (2017, p. 2) for references to this literature.

[6] The former point is of notable interest, not the least given the public finance concerns that motivated the downscaling of German and Spanish subsidy programs, see Böhringer et al. (2014). Japan introduced in 2012 a feed-in tariff of about 0.41 EUR per kWh (Ma et al., 2015, p. 87) .

[7] A cap-and-trade system that caps CO_2-emissions within the European Economic Area (EU plus Iceland, Liechtenstein, and Norway).

There are other arguments (e.g. capital market imperfections that reduce investment in a benign technology), that all point to a deviation between the privately and socially optimal level of investment in "green power". Many of these arguments are all based on the presumption that there is a positive externality related to innovation. Yet, it is not easy to find this in the data. For example, with respect to the Swedish tradable green certificate scheme, Bergek and Jacobsson (2010, p. 1266) conclude that substantial rents are reaped by investors but the scheme has failed to stimulate early nursing and bridging markets. Our empirical analysis gives, at least indirectly, support to this finding, in that the main economic effect of *phasing out* the certificates will be a transfer from certificate owners to users of electricity that must buy certificates. Lehmann and Gawel (2013) contend, however, that combining a green certificate scheme with a CO_2-emissions cap-and-trade program help to unravel technological gains that otherwise would not have materialised. This argument seems to underestimate the possibility of technological spillovers available if other activities in the economy were subsidized instead (as also pointed out by e.g. Schmalensee, 2012).

Perhaps, the most popular argument related to externalities is the possibility to reduce negative externalities from e.g. fossil-fueled electricity generation, especially when political economy considerations preclude Pigou pricing. In particular, proponents argue that subsidies to green energy has helped to improve local air quality, typically via the shutting down of coal plants. In the next section, we will link this argument to studies that have tried to quantify these gains. For now, we just mention an important caveat pointed out by Schmalensee (2012, p. 48), namely that adding renewable power plants is more likely to displace plants with high marginal costs, like gas-fired ones, rather than coal-fired plants.

Even if a first-best instrument is in place to mitigate a particular emission, it is still possible that other emissions may be reduced by a shift to green technologies. Such "co-benefits" have been highlighted by e.g. Meyer et al. (2014, p. 14). Of course, if these emissions are correctly taxed, the externalities are already internalized — there are no net benefits. The Swedish sulphur-tax is an (imperfect) example. It is paid, inter alia, by fossil-fueled electricity generators, who also must possess carbon permits under EU-ETS.[8]

Finally, there are negative externalities involved per se, as wind power may cause noise and other problems for human beings, and adversely affect birds and bats. There are also claims that wind farms can alter the nearby rainfall and temperature; refer to, for example, Zhou et al. (2013).

2.2. Jobs

Support to green technologies have been argued to promote employment — they can "create green jobs".[9] The "green jobs" argument must be considered in relation to how jobs are "created" in alternative scenarios. For example, if a country rather imports electricity than support a particular type of domestic electricity generation, there will be adjustments in relative prices and exchange rates. There must necessarily follow an increase in exports to pay the imports; domestic production displaces imports to clear the balance of trade. These activities can also be said to "create" jobs. In short, whatever the "job creation" potential of a subsidy scheme might be, it must be compared to an alternative. Increases in end-user electricity prices caused by green certificates could also hamper employment in various parts of the economy. Lyon and Yin (2010) find that US states with high unemployment rates are slower to adopt green certificate schemes (renewable portfolio standards). Boampong et al. (2016) undertake an ex post analysis of the impact on employment of US green certificate schemes. They find that the schemes have had no significant impact on overall employment. In brief, it is not easy to document a "green jobs" effect, the uncertainties seem pervasive. Indeed, in their survey of this topic, Cameron and van der Zwaan (2015, p. 160) "observe significant uncertainties in quoted figures for job creation, both across and within publications".

2.3. Energy security

A third argument is related to "energy security", a term that is not well-defined, but may refer to the degree of self-sufficiency. One argument for increasing (green) domestic electricity production is to make a country less dependent on imports. This is claimed to increase energy security, although such an argument is hard to evaluate in the abstract; its validity depends inter alia on the specifics of the energy market under consideration.[10]

Another energy security-related problem deserves mentioning. Imbalances in the physical trade on the electricity spot market must be levelled out in order to maintain the balance between production and consumption, and to maintain power grid stability. When more intermittent electricity is added to the grid, grid stability problems are more likely to arise. Thus, with large penetrations of intermittent sources, one would expect the power system to face some extra balancing costs.[11] Using a numerical approach, Hirth (2016) finds that the market value of electricity from wind declines with penetration, but it tends to decline at a slower rate if flexible hydropower is present, as in the Nordic Nord Pool market, than in markets dominated by thermal power plants, such as in Germany.

This concludes our discussion of the arguments in favor of subsidizing "green power". In the next section, we review some empirical studies and begin with multicountry studies, after which we proceed to studies that look at a system within a given country.

3. Empirical work: a brief survey

Perhaps the most extensive study on the benefits of switching to a system completely based on renewable energy was carried out within an OECD project on greening household behavior, see OECD (2013). Using two large-scale waves (2008, 2011) of surveys covering 21 countries[12], the benefits were measured by a contingent valuation style question. It asked for the value (in terms of a % increase of the electricity price) of a complete switch to renewables in the respondent's own country, given the household's current level of electricity usage. While the variation of valuations is significant, both between and within countries, but to a lesser extent across the two surveys, willingness-to-pay was found to be in the order of 5–10 % increase in the electricity bill. This is quite consistent with how much of an extra payment energy companies are able to extract from "green power" contracts (OECD, 2013). It should be stressed, however, that the OECD-effort was not intended as a cost–benefit analysis of renewable energy policy options, nor was the valuation part intended to mimic a state-of-the-art contingent valuation study. Kelsey and Meckling (2018) compare ownership of renewables in 41 US states and 18 EU states, trying to disentangle (what they term) "winners" and "losers" in terms of ownership type. They find that

[8] The example is imperfect in the sense that the sulphur tax may not be correctly set, but since it is positive, one must be aware of double-counting the value of the environmental co-benefits.

[9] There is also the question of how to value employment. Setting the reservation wage of (unemployed) labor equal to zero may overestimate the value of alleged job creation; the reservation wage is positive as long as leisure has a value.

[10] For our example, Sweden, this seems to be a marginal issue, because the country is part of the Nordic Nord Pool electricity market. Roughly every second year, Sweden is a net exporter/importer of electricity. The annual net import/export typically is less than 1.5 TWh, compared to a yearly consumption of around 140 TWh.

[11] For Sweden, grid energy storage is not considered to be an economically viable option within the foreseeable future.

[12] In total, some countries were sampled only once.

independent power producers are "winners" regardless of the particular support policy used to stimulate renewables. We next turn to a few single-country studies.

Bergmann and Hanley (2012) provide a comprehensive study from Scotland, a country that has rapidly expanded its portfolio of renewable energy generators. The study includes a comprehensive review of the relevant literature on costs and benefits. In particular, they scrutinize a range of studies on (mainly) the environmental costs and benefits associated with hydropower, off- and onshore wind and biomass. The subsidies may provide a boon to rural areas and to economic growth in general. At the same time, the subsidies to renewable energy in the UK has increased consumer electricity bills. The rising bills has contributed to a vigorous debate on "fuel poverty" not only in the UK, but also in other EU-countries.

Turning to Germany and its flagship Energiwende program, we first note that the program has been scaled down in recent years. This appears to be consistent with Andor et al. 's (2017) finding that Germany's willingness-to-pay (WTP) for green electricity is falling over time. Böhringer et al. (2014) look at alternative ways of financing the Energiwende. The authors report that the feed-in tariffs totalled 21.1 billion EUR in 2014, propelling the revision of the system to add more flexibility and lower remunerations. A CGE-model is combined with micro-data, to study distributional effects of alternative financing schemes. Of the options studied, broad-based VAT-financing seems to be the preferable way (in welfare terms) to fund the subsidy. Abrell et al. (2017) argue that an expansion of renewable energy has two distinct effects on the electricity market, a replacement effect and a price effect. First, conditional on the price, conventional energy generation is replaced by renewable energy, potentially reducing emissions from fossil fuels. Second, the increase of supply reduces the market price and pushes higher marginal cost technologies out of the market. Using an econometric approach, they study the cost (per unit of CO_2) of subsidy programs in Germany and Spain, finding significant variation of the cost of reducing a ton of carbon across renewable technologies (and the two countries). A particularly interesting finding implies that consumers are better off with a feed-in tariff (compared to no-tariff); the lower electricity price makes up for the cost of financing the subsidy (through an electricity tax). The existence of EU-ETS means that the total carbon emissions are given, so that the calculated "…*cost of carbon abatement through the RES policy per ton of* CO_2" (Abrell et al., 2017, p. 10) is difficult to interpret in a cost–benefit sense (one could argue that the cost of the policy is infinite per unit of CO_2 avoided in the atmosphere).

Next, we provide two Nordic examples. Bye (2003) evaluates the social benefits and costs of a hypothetical Norwegian green certificate scheme. At a green share of 10%, he estimates the social loss to amount to around NOK 1 billion (or about 0.1 billion EUR per year). Johansson et al. (2016) consider an increase in the Swedish green certificate quota by one tenth of a percentage point (from 23.1 to 23.2% in 2016). They calculate the annual social cost of this marginal increase in the quota to be SEK 17 million (around EUR 1.60 million).

A substantive assessment carried out in the US is reported in Wiser et al. (2017). They find that the total opportunity costs of the certificates, in the 29 states where they are used, is about 31 USD billion. This sum is, however, found to be outweighed by environmental benefits, co-benefits in particular. The authors stress the point that existing certificate-based systems are not the first-best solution to environmental issues, but the co-benefits produced could be sufficient to pass a cost–benefit test.

Cullen (2013) studies wind power offsets for generators on the Texas electricity grid. He suggests that the value of subsidizing wind power is driven primarily by CO_2 offsets. However, the social costs of CO_2 would have to be greater than USD 42 (around EUR 35) per ton in order for the environmental benefits of wind power to exceed the costs of subsidies. In a follow-up study, Novan (2015) uses detailed data about wind power production in Texas. He claims that the subsidized plants contribute "substantially" to emission reductions of several pollutants. He also explains that the benefits vary because demand varies over the 24 hour day; different technologies are operative at the margin. Consequently, a flat subsidy rate is not as efficient as one that would take the temporal differences into account. Rouhani et al. (2016) look at the costs and benefits of various 2020 California targets of renewable portfolio standards. Under almost every scenario evaluated, the mandated 33% green certificate share does not offer the highest social welfare benefits.

The Canadian province of British Columbia has a green certificate scheme known as the clean electricity standard. Rhodes and Jaccard (2013) find that the clean electricity standard will reduce carbon emissions four to six times more than the carbon tax. The average cost per ton of CO_2 reduced is significantly higher than the carbon tax.

Having briefly reviewed some of the empirical evidence, we now turn to our model and its implementation. We begin in the next section, by deriving a general equilibrium cost–benefit rule for arbitrary changes in a quantity-based subsidy scheme. This model is then taken to the data in Section 5.

4. A general equilibrium cost–benefit rule

We use a simple social welfare function, implicitly based on a general equilibrium model, to derive a cost–benefit rule that can be used for the evaluation of certificates (as well as other policy measures). This rule is applicable for the evaluation of small as well as large changes in green certificates. In order to focus on the green certificate system, while setting aside distributional issues, we consider a single representative household. There are three commodities: electricity, an aggregate composite commodity, both subject to at least a value added tax (VAT), and an untaxed numéraire.[13] Given equilibrating markets there is no need for further disaggregation, because small price adjustments induced by the considered project will have no welfare implications, as is illustrated by Eq. (9) in Appendix A.1.

The indirect utility function of the representative household also serves as the social welfare function:

$$V = V(\mathbf{p}^f, \mathbf{p}^{Af}, y + \pi + \pi^A + T, \mathbf{E}), \tag{1}$$

where $\mathbf{p}^f$ and $\mathbf{p}^{Af}$ denote $1 \times \tau$ vectors of present value final user electricity prices and final (consumer) composite commodity prices, respectively, τ denotes the time horizon, $\mathbf{E}$ denotes a $1 \times \tau$ vector of emission levels, y denotes a present value lump-sum income, $\pi = \pi^G + \pi^{NG}$ denotes present value profits of green plus non-green power plants, π^A denotes present value profits in the composite commodity sector, and T denotes a (positive or negative) present value payment from/to the government. One could view $\mathbf{E}$ as a function of all relative prices. Provided the considered country is part of a tradable permit system for a gas, such as the EU ETS for CO_2, policies will have no impact on total emissions of the gas: the provision of permits is exogenous so that domestic policy changes may affect the permit price but not the total number of permits, as long as the permit price is strictly positive. Thus, in such a case, $\mathbf{E}$ can be viewed as a shortcut for emissions of harmful gases/particulates, except any gas that

[13] In the kind of models considered here, only relative prices can be determined. This explains that the price of one commodity (denoted the numéraire) is normalized to unity. Refer to any textbook on microeconomics, e.g., Jehle and Reny (2011, p. 50), for details. We assume that the supply of the numéraire is infinitely elastic, and that it is used to produce the other commodities.

is subject to a cap and trade system. Refer also to the discussion in Section 6 on national versus global levels of emissions.

The time t quota, denoted α_t, requires that a certain proportion of electricity is produced by green power plants:

$$\alpha_t = \frac{e_t^G(p_t^s + p_t^c)}{e_t(\mathbf{p}^f, \mathbf{p}^{Af}, y + \pi + \pi^A + T, \mathbf{E})}, \tag{2}$$

where $e_t^G(.)$ denotes the supply of green electricity in period t, p_t^s denotes the spot price of electricity, p_t^c denotes the green certificate price, and $e_t(.)$ denotes the period t demand for electricity by those who must possess green certificates. Note that Eq. (2) is an equilibrium condition. It stipulates that in equilibrium the certificate price is such that α_t percent of aggregate electricity is supplied by green producers.

The monetary welfare impact of a change in the green certificate scheme, that is, $\Delta\alpha_t \neq 0$, can be modeled as a change in the certificate price from its actual general equilibrium level in period t to a new level. This causes changes in other prices and incomes. A general equilibrium cost–benefit analysis of the policy move can compactly be stated as follows:

$$V(\mathbf{p}^{f1}, \mathbf{p}^{Af1}, y + \pi^1 + \pi^{A1} + T^1 - CV, \mathbf{E}^1) = V(\mathbf{p}^{f0}, \mathbf{p}^{Af0}, y + \pi^0 + \pi^{A0} + T^0, \mathbf{E}^0), \tag{3}$$

where a superscript 1 (0) refers to general equilibrium levels after (before) the change in the quota, and CV denotes the compensating variation. This is the maximum once-and-for-all willingness-to-pay (WTP) for the considered change in the certificate scheme. If negative, CV is interpreted as the minimum compensation (WTA) needed in order to be as well off without as with the considered change in the certificate scheme. General equilibrium is here interpreted in the conventional way, that is, as signaling that relative prices are flexible, not as indicating that we account for every conceivable effect of the considered policy measure.

Consider now an evaluation of the compensating variation. In Eq. (8) in Appendix A.1, we state the resulting general equilibrium cost–benefit rule (a line integral). Here, we focus on a simplification of the rule that is easier to compute:

$$CV \approx -\sum_{t=1}^{\tau}[\Delta e_t \frac{\Delta p_t^f}{2} + e_t^0 \cdot \Delta p_t^f] + \Delta\pi + \Delta T + \sum_{t=1}^{\tau} \frac{V_{E_t}}{V_y} \Delta E_t, \tag{4}$$

where $\Delta e_t = e_t^1 - e_t^0$, $\Delta p_t^f = p_t^{f1} - p_t^{f0}$ denotes the period t change in the present value end user price, $\Delta\pi$ denotes the change in present value profit income of power plants, ΔT denotes the present value change in tax income, V_{E_t} denotes the time t marginal disutility of emissions, V_y denotes the marginal utility of present value income, and the time t change in emissions is denoted ΔE_t. Eq. (4) is exact if preferences are quasilinear and the demand function for electricity is (locally) linear in p_t^f, otherwise the two first terms represent a linear Marshallian consumer surplus approximation of the Hicksian WTP measure of a price change; refer to Eq. (8) in Appendix A.1. [14] The first term within brackets provides the period t consumer surplus triangle; see Fig. 2. The second term is the change in period t consumer surplus on the initial (final) e_t^0 units if the price decreases (increases). Taken together, these two terms reduce to the rule of half. The WTP/WTA for a change in present value lump-sum income equals the change itself, explaining the third and fourth terms. The sum of the four first terms in Eq. (4) reduces to the sum of present value consumer surplus triangles plus the change in present value tax revenue ΔT. By assumption, the change in pollution is so marginal that point estimates of V_{E_t}/V_y can be used to approximate the WTP/WTA in exchange for small changes in pollution, as is done in the final term of the equation.

In closing this section, we stress that Eq. (4) can be applied to marginal changes in a certificate scheme as well as large changes, including discarding a scheme. In the limit as $\Delta\alpha_t$ goes to zero, Eq. (4), as well as Eq. (8) in Appendix A.1, reduces to the sum over t of $-p_t^c \cdot e_t^0 d\alpha_t$ plus dT; refer to Dietz and Hepburn (2013) for a comprehensive analysis of the shortcomings of marginal cost–benefit rules in evaluating large policy changes. In fact, with no or only minor adjustments our evaluation rule can be used to assess any policy measure that impacts on prices, taxes, and environmental parameters. Thus, Eqs. (4) and (8) in Appendix A.1 provide general and empirically meaningful evaluation rules.

5. Application to Sweden

The Swedish electricity certificate scheme was introduced on 1 May 2003 and initially planned to end on 2035. The purpose of the system is to increase generation of certified sources. Not all renewable sources are included, e.g. large-scale hydropower.[15]. Since 1 January 2012, Sweden and Norway have shared a common green certificates market. The goal is to increase the production of renewable electricity by 28.4 TWh by 2020, relative to 2012, or 10% of the two countries' combined electricity production. In addition, Sweden has its own target (Parliament decision 1 June 2017) of increasing renewable electricity production by 18 TWh to 2030. The decision entails closing the scheme in 2045. Fig. 1 illustrates the quota curve 2003–2045.[16] For example, in 2019 electricity consumers required to possess certificates have to purchase 28.8% of their electricity from "green power" plants that are awarded certificates. Given the focal point of the paper, we ignore the Norwegian participation in the system.

Before presenting the empirical results, we detail a few assumptions about the electricity market under consideration.

5.1. Basic assumptions

The supply of electricity on the Nordic spot market is typically viewed as a merit order curve, where different types of plants have different but constant marginal costs. Equilibrium is reached where the demand curve intersects the merit order curve. This is illustrated in Fig. 2. The difference between the spot price of electricity and the final (Swedish) consumer price is made up of the cost of certificates, variable transmission tariffs, the energy tax, and VAT.

All plants are assumed to have constant marginal costs up to capacity. Moreover, assume that the total capacity of green power plants is the same with and without green certificates; it is worth noting that Upton and Snyder (2017) find no evidence that

[14] If $\Delta\pi + \Delta T \neq 0$, the Marshallian demand curves shift, unless income elasticities are zero. Drawing on Kriström (2012), we expect the income elasticity of demand for electricity to be small, at least for many OECD-countries, including Sweden.

[15] Eligible for certificates are certain hydro, certain biofuels, geothermal, solar, peat in CHPs and wave energy installations.

[16] The revision decided in June 2017 actually changed the requirement from a proportion to a quantity (TWh), the required proportion is then backed out ex post from observed market data and includes an adjustment mechanism.

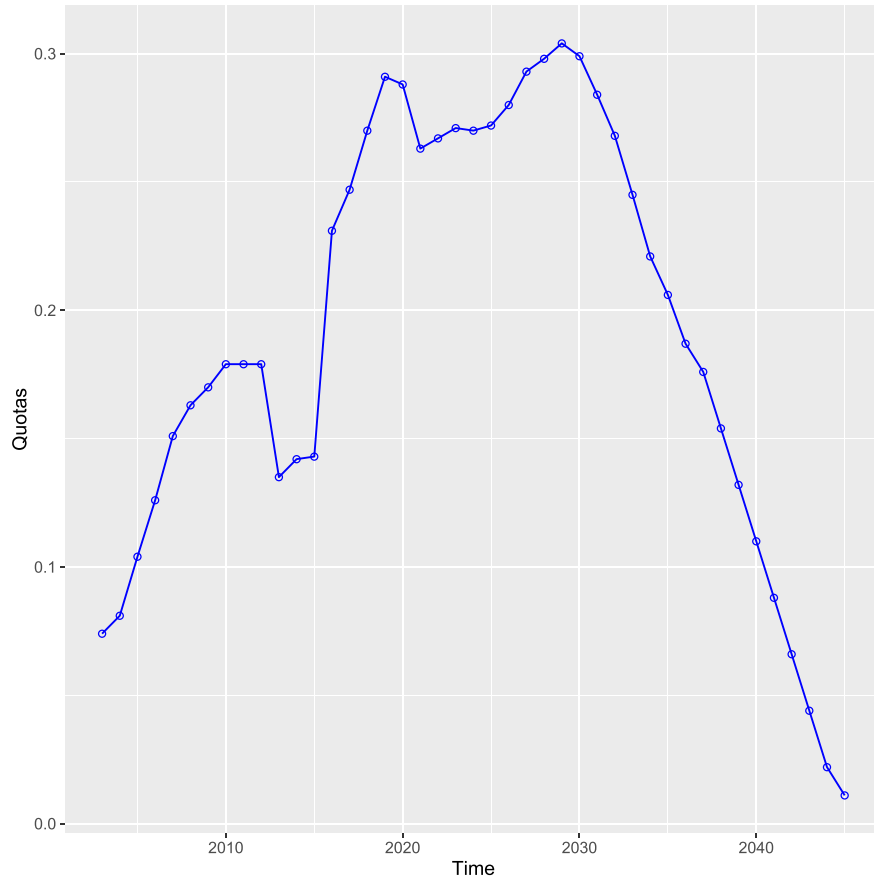

Fig. 1. Swedish certificate quotas 2003–2045. .
Source: Swedish Energy Agency (2018b)

certificates (RPSs) are associated with increases in renewable electricity generation in the US. Finally, the certificate price and the end user price are allowed to change, while all other prices are kept constant (or marginal changes are allowed as in Eq. (9) in Appendix A.1). As a consequence, and as in Fig. 2, the same type of power plant is marginal in both considered scenarios.

Consumers have to pay the spot price of electricity plus $\alpha_t \cdot p_t^c$, that is, the required proportion of green electricity times the certificate price (per kWh throughout). In addition, consumer pay transmission fees[17], an energy tax and a value-added tax. Therefore, the Swedish final user electricity price (for consumers that need certificates) is defined as follows:

$$p_t^f = (p_t^s + \alpha_t \cdot p_t^c + t^e + p_t^d) \cdot (1 + t^m), \tag{5}$$

where p_t^f denotes the final time t consumer price, p_t^s denotes the spot price, α_t denotes the quota, p_t^c denotes the certificate price, t^e denotes a unit tax, p_t^d denotes the variable transmission tariff, and t^m denotes the value-added tax.

We also need to specify the government's budget constraint. For simplicity, we focus on taxation of the electricity sector and the composite commodity, while all other activities by the government are suppressed:

$$T_t = [t^e + (p_t^s + \alpha_t \cdot p_t^c + t^e + p_t^d) \cdot t^m] \cdot e_t(.) + t^m \cdot p_t^A \cdot e_t^A(.), \tag{6}$$

where $e_t(.)$ denotes final electricity demand (by consumers required to have certificates), p_t^A denotes the producer price of the composite commodity, and $e_t^A(.)$ denotes demand for the composite commodity.

Swedish Energy Agency (2016, p. 86) forecasts that Swedish electricity demand requiring certificates will be around 90 TWh per annum to 2035, while total electricity demand will be around 140 TWh; energy-intensive industry, which is exempt from certificates, uses around 40 TWh, while transmission losses

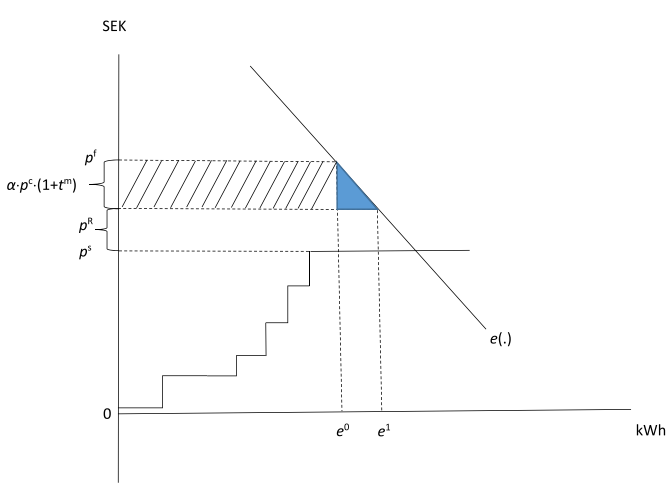

Fig. 2. A stylized picture of the Nordic (Nord Pool) spot market, where p^R denotes the residual.

[17] In a regulated natural monopoly, customers often face a two-part tariff, consisting of a fixed monthly charge (here deducted from lump-sum income) and a tariff per kWh.

amount to about 10 TWh. We will use these numbers throughout our calculations.

The empirical parts of the paper are based on a data set collected during 2017 and early 2018 for a by now completed part of a research project (sponsored by Eforis). This explains why we will make a distinction between the periods 2003–2017(ex post) and 2018–2045(ex ante), respectively. Next, we turn to the first part of our empirical assessment. In the ex post analysis, we look back at observed market values during 2003–2017, while the ex ante analysis is the more conventional CBA, in which we look at costs and benefits in the future (of scrapping the system 2018, rather than running it to 2045).

5.2. Ex post analysis of 2003–2017

A cost–benefit analysis is typically carried out with the purpose of supplying information about an investment. Nevertheless, an ex post calculation within the project period can provide useful insights about the policy's effectiveness. We thus begin by looking at the benefits and costs of having the system in place during 2003–2017.

If the certificate system is phased out, the price of electricity could be affected as per Eq. (5), since the subsidy no longer has to be paid. To calculate the ensuing effect on electricity demand, we set the price elasticity to −0.3(a consensus estimate according to the survey in Kriström (2012), it is almost certainly lower (in absolute value) in the short-run and higher in the longer run) and solve:

$$\frac{\Delta e_t}{\Delta p_t^f}\frac{p_t^f}{90\cdot 10^9} = -0.3. \tag{7}$$

for 2003–2017, using the observed consumer electricity prices. We approximate the counterfactual electricity consumer price by putting $\alpha_t = 0$ for all t in Eq. (5). This simplifying assumption is supported by extensive econometric analysis of the Nordic electricity — certificate markets in Schusser and Jaraite (2018) . However, to obtain a robustness check, in the sensitivity analysis, we draw p^s and p^c from their observed distributions.

In the welfare analysis, we first focus on the electricity market and set aside environmental concerns and the induced effects on final demand. Thus, we compute consumer surplus and profit changes and add the changes of tax revenue (energy tax plus VAT) as pertaining to the electricity market, using Eqs. (4), (7) and (6). The data for this calculation is reported in Appendix A.3, seeTables 2 and 3. We compute the capitalized value using 2018 as base year, using two different discount rates. The first one, equal to 1.4%, is taken from the Stern Report, see Stern (2007). The second one, equal to 3%, is taken from the EU manual on CBA, see European Commission (2014).

We follow the recommendation of European Commission (2014) regarding the backwards discounting procedure (present values are computed with time running from −15 to −1, i.e. 2003–2017, since we use 2018 as the base). This means that the capitalized values will be increasing in the discount rate, in contrast to the usual present values. This should be remembered when looking at the results.

The complete ex post CBA is in the left panel of Table 1, denoted Table 1a, but we begin discussing the effects arising at the level of the electricity market. On the consumer side, we find lower prices and a consumer surplus gain via increased use; the losses made on the producer side due to lower prices means lower income from profits — the single representative household owns all firms. An increased electricity demand increases revenue from the per unit energy tax, but this is counteracted by a loss of VAT income. The usual tax elasticity formula will hold, so we expect to find a decrease of tax revenue, given an inelastic demand for electricity.

Table 1
CBA results. ΔCS = consumer surplus change, $\Delta\pi$ = change in electricity sector profits, ΔT^{el} = change in electricity sector tax revenue, CV^{el} = net sector benefits, ΔT^c = change in VAT tax revenue, composite sector, CV^E = externality cost, CV = Economy-wide net benefits. Note: results above the horizontal line in panels (a) and (b) pertain to the electricity market, those below to the whole economy.

	Discount rate 1.4%	Discount rate 3%
(a) Ex post 2003–2017 in 2018 billion SEK		
ΔCS	60.13	68.56
$\Delta\pi$	−47.85	−54.56
ΔT^{el}	−4.32	−4.93
CV^{el}	7.96	9.07
CV^E	−1.07	−1.22
ΔT^c	−1.93	−2.20
CV	4.96	5.65
(b) Ex ante 2018–2045 in 2018 billion SEK		
ΔCS	38.62	33.31
$\Delta\pi$	−30.82	−26.57
ΔT^{el}	−2.31	−1.99
CV^{el}	5.50	4.74
CV^E	−0.67	−0.58
ΔT^c	−1.19	−1.03
CV	3.63	3.13

5.3. Ex ante analysis 2018–2045

We now turn to a forward-looking calculation, using the same methodology as in the previous section. We need to make a series of assumptions about future prices, quantity and tax parameters. The precise assumptions used are listed in Appendix A.3, in Table 3. The future electricity price we use for the base calculation is consistent with the long-run forecasts produced by the Swedish Energy Agency, which is lower than the average of 2003–2017. For this ex ante analysis, we use the observed average certificate prices during 2017, which are substantially lower compared to the previous years when the scheme has been running. The prices we use arguably yields a conservative estimate of the value of scrapping the scheme. We contrast this in the sensitivity analysis, where we instead draw prices from the observed empirical distribution. In this way, we obtain an additional robustness check of our analysis.

The tax parameters are set to 2017 values and we simply assume that there will be no future changes. The quantity demanded with the certificate scheme in operation is assumed to be 90 TWh, as per the discussion above. As in the ex post case, we first discuss the results without explicitly taking into account larger economy-wide repercussions and externalities, see the right panel of Table 1, denoted Table 1b. The results from the ex ante analysis limited to the electricity market are qualitatively similar to the ex post analysis. The main effect is a profit loss and ensuing increase in consumer surplus; as before, this is essentially a transfer between certificate owners and consumers. While qualitatively similar, the results show markedly lower surplus gains and losses. The reason is, as noted, that we use significantly lower prices.

5.4. The complete CBA

The results discussed so far are perhaps not surprising, since a subsidy will necessarily lead to welfare losses in the aggregate in a first-best world (barring any distributional concerns). As shown above, the net gain of a small reduction of this particular subsidy scheme can be (very roughly) approximated by the certificate price times the reduction of electricity generation covered by certificates. By construction, this measure is positive (note that it is still a general

equilibrium result). Hence, the standard intuition has some bearing on this case, even though the structure of certificates as a subsidy scheme is quite different from, say, a unit subsidy. However, as Eq. (4) shows, there are additional items to consider in a complete CBA of certificate schemes, i.e. economy-wide public finance repercussions and externalities. These items arise because we are in a second-best world.

The phasing out of the certificate scheme causes disposable lump-sum income received by the representative household to fall; both $\Delta\pi^G$ and ΔT are negative. Because the composite commodity is assumed to be a normal good, the loss of lump-sum income causes the associated demand curve to shift to the left. In turn, this shift of the demand curve may cause a loss of VAT-income. As an extreme case, suppose that the aggregate supply curve for composite commodity is completely elastic and that there are no savings or spending abroad. Then, roughly, the loss of VAT is covered by the loss of disposable income less the change in expenditure on electricity multiplied by the average VAT-rate (estimated at 20%).

As illustrated by the final term of Eq. (4), consumers may value a shift from fossil fuels to green energy. The meta-regression analysis of the extra or additional WTP for green electricity by Ma et al. (2015) suggests a mean for this extra WTP equal to SEK 0.0088 per kWh; refer to Appendix A.2. However, it is difficult to judge if this is a reasonable estimate for Sweden. Therefore, in the absence of any direct estimates of the Swedish WTP (WTA) for using less (more) fossil fuel-based electricity, we base our estimates on the online calculator Ecosense LE (ecoweb.ier.uni-stuttgart.de/EcoSenseLE). On average, a coal-fired power plant is considered to be marginal in the Nordic (Nord Pool) market. According to the estimates reported in Appendix A.2, a reasonable *upper limit* for the environmental and health costs is SEK 0.07/kWh. Using this as an upper bound, and, as an illustrative example, phasing out the current system in the beginning of 2018, this externality, denoted CV^E, has a present value cost of about 0.67 (0.58) billion SEK at 1.4% (3%).

Expressed as present values for the period 2018–2045, we estimate the loss of VAT income in the composite sector, denoted ΔT^c, to be about SEK billion 1.19 (1.03) when the discount rate is 1.4 (3)%. If we add those values, the total CBA gives in the ex post calculation a net value of 4.96 (5.65) SEK billion, and in the case of the ex ante 3.63 (3.13) SEK billion for the two discount rates used. Thus, adding these two second-best dimensions to the analysis will reduce the total benefit, denoted *CV*, from removing the scheme. An in medias res analysis sums the ex post and ex ante analyses to obtain net benefits of phasing out the certificates equal to 8.59 (8.78) SEK billion. Because we have used parameter values that allegedly are not biased against the certificate scheme, rather the opposite, these results seem quite robust. We will buttress this conclusion by a sensitivity analysis below.

Before we turn to our sensitivity analysis, let us note that in an otherwise undistorted economy, it is straightforward to show that maximizing Eq. (1) with respect to α results in an optimal α that is equal to zero (in each period). However, in a second-best world with distorting taxes and pollution, our results show that this is no longer necessarily true; reducing α causes both benefits and costs. However, it seems more or less impossible to empirically estimate the optimal α-path over time. Therefore, a CBA based on a comparison of the actual α-scheme and $\alpha = 0$ seems to be the best available option.

5.5. Sensitivity analysis

There is a significant uncertainty about many of the parameters we have used in the ex ante analysis. We therefore include a systematic sensitivity analysis by means of a simple Monte-Carlo analysis. As noted above, we use different market prices for the ex ante simulation, which in itself is a kind of sensitivity analysis. For the future prices, we use the observed distribution 2003–2017 of electricity and certificate prices and estimate the best fitting distribution. The price distributions fitted in the simulations are displayed in Fig. 4 in Appendix A.4, where the estimation procedure is described. We have little information about the distribution of the environmental values, and therefore take them to be uniformly distributed; the bounds are based on the empirical studies reported above, see Appendix A for details. To allow for the possibility that the associated positive and negative externalities "net out", we include zero as the lower bound.

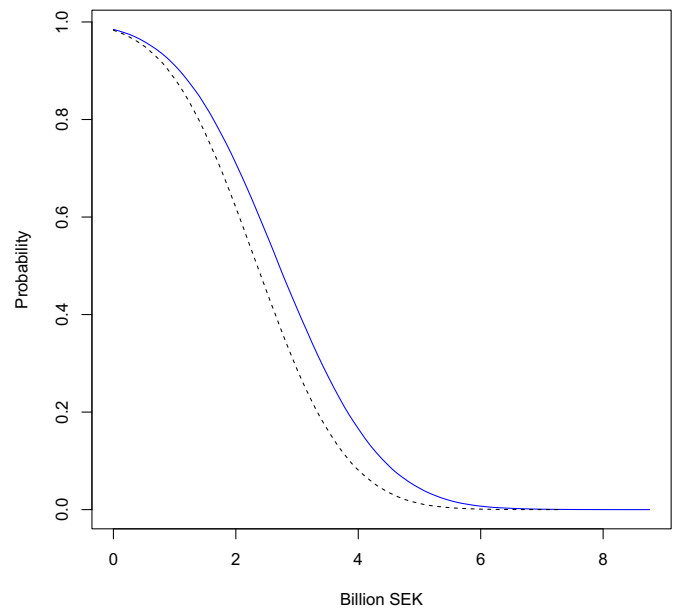

Fig. 3. Estimated cost-benefit acceptability curves when the discount rate is 1.4% and 3% (- - -), respectively.

Importantly, we assume independent draws[18] and use a Monte-Carlo analysis with repeated draws from the three distributions used. The present value in each simulation is thus obtained to compute the empirical survival distribution (i.e. 1 minus the empirical c.d.f.). Johansson and Kriström (2016) call this curve the "cost–benefit acceptability curve", because it provides a concise summary of the project's potential profitability. The curves for the two considered discount rates are shown in Fig. 3. To illustrate the interpretation of such a curve, the probability is around 0.99 that the outcome of the CBA for the period 2003–2045 is non-negative, i.e., there is just a slim chance that the certificate scheme is socially profitable. The probability is 0.7 that abstaining from the certificate scheme would have produced a social present value gain that exceeds SEK 2 billion when the discount rate is 1.4%.

This Monte-Carlo analysis suggests that the basic results are quite robust towards perturbations of the underlying assumptions on certain key parameters. It requires a rather extreme combination of price and externality values for the present value to be negative, but an upward valuation of the externalities will increase the likelihood that the scheme passes a cost–benefit test. Yet, our calculations are based on estimates garnered from the literature that do cover a wide range of externality costs.

[18] The conventional correlation-coefficient between p^s and p^c is 0.504 in 2003–2017, suggesting a linear bivariate dependence. Our independence assumption is based on the econometric findings of Schusser and Jaraite (2018). Other outcomes are reported by Fridolfsson and Tangerås (2013), Helgesen and Tomasgard (2017), and Liski and Vehviläinen (2016).

6. Conclusions

The approach developed in this paper should be useful for evaluations of changes in a green certificate scheme in a general equilibrium setting. We illustrate the framework by analyzing the social benefits and costs of phasing out the Swedish green certificate scheme at the beginning of 2018, together with an ex post assessment for the 2003–2017 period.

A more extensive welfare analysis of the reform will also include a scrutiny of electricity system effects and distributional concerns. Due to lack of data, the power grid stability issue has been omitted. If cancelling the green certificates causes the penetration of intermittent electricity to decrease, one would expect a less volatile spot price of electricity. However, there seems to be no consensus with respect to the severity of the "balancing problem" in the Nordic power market, not even among Swedish energy organizations and corporations. Nevertheless, Greenstone et al. (2019) claim that intermittency costs and costs of added grid infrastructure are important cost drivers of certificates (RPSs) in the US. There are also important distributional consequences of green certificates. Apparently, the green certificates cause considerable redistribution from electricity consumers to owners of green power plants. If the certificates are phased out, consumers gain through the lower electricity price, a gain that is increasing with electricity consumption. Given a low price elasticity of electricity, the change is likely to be progressive, although we have no detailed information about the exact distributional effects on the household side.

Furthermore, we have also ignored any (positive or negative) employment effects of green certificates. Sweden faces a shortage of skilled laborers, and one expects that investments in green technologies will displace other activities. In addition, higher end user electricity prices caused by the green certificates are likely to hurt employment in sectors that are required to possess certificates. Conversely, if the certificates are removed, there is a loss of tax revenue, that somehow must be compensated at a given size of the public sector. The net environmental effects of the suggested reform are probably negative, although information is scarce. Under our assumptions, there will be more local fossil-based emissions (but no additional carbon emissions), in Denmark and/or in Sweden. On the other hand, with the likely dismantling of wind power if the subsidy scheme is scrapped, we will also find positive ecosystem effects.

We have assumed that green power plants will be built even in the absence of certificates. This assumption (by the way, fitting US data according to Upton and Snyder, 2017) yields a reasonable lower limit for the social profitability of abstaining from certificates. In future research, we plan to use micro-simulation and computable general equilibrium models to shed further light on this issue. This could help understanding of how system design affects the effectiveness of certificates.

There are obviously policy issues involved that deserve discussion. If the intention was to reduce harmful emissions of greenhouse gases, the certificates were "overrun" by the EU ETS. Nevertheless, as pointed out by a reviewer, policy makers may have targets for domestic emissions. However, it is hard to put a price tag on measures that reduce domestic emissions but have zero impact on global emissions, i.e., merely reshuffles emissions across countries and hence aggravates attempts by other countries to reduce their emissions. In any case, it seems as the ambition rather has been to achieve (over time more and more ambitious) targets with respect to the number of TWh supplied by certain types of renewable sources (wind, solar, and so on). Our analysis assumes that the time pattern for the addition of renewable sources would have been the same in our two scenarios (realized versus alternative). As argued previously, this assumption provides a kind of lower bound for the (absolute value of the) social loss of introducing the certificate scheme; adding the kind of balancing costs identified by Greenstone et al. (2019) would considerably increase the lower bound for the loss. A more realistic assumption with respect to the alternative scenario is as follows. In the absence of certificates, the buildup of renewable power plants would have been slower than in the observed scenario. The reason for this claim is a quite fast – and from the small open economy's point of view exogenous[19] – technological development. Over time, this development has lowered variable as well as fixed costs of wind farms and other renewable power plants. In fact, newer plants are competitive even without subsidies. Therefore, in the alternative scenario, we would most likely have ended up with about the same installed capacity as the one observed.

However, investing at later points in time and using cheaper technologies would have saved costs to society. This claim is supported by recent proposals to "close" the certificate system, i.e., not allocating certificates to plants becoming operational beyond a certain point in time.[20] Without such a stop mechanism, the fear is that the certificate price will become so low that older high-cost plants go bankrupt. A somewhat counter-intuitive consequence is that the certificate system possibly has achieved a target for installed TW of renewable alternatives, but also has shifted investments from later and cheaper to earlier and more expensive technologies.

One could possibly argue that there are ethical reasons for contributing to an international journey towards a sustainable world. On the other hand, there are also strong reasons for economizing on taxpayers' money. However, providing advice on such ethical issues is beyond our area of competence. Our more modest mission has been to apply economic theory to improve the decision basis for the selection of policy instruments in the future.

A.1. Theory

Using Eq. (3), we arrive at the following general equilibrium cost–benefit rule:

$$CV = -\sum_{t=1}^{\tau}\left[\int_{p_t^{f0}}^{p_t^{f1}} e_t^H(.)dp_t^f + e_t^{AH}(.)\cdot(1+t^m)dp_t^A\right] + \Delta\pi + \Delta\pi^A + \Delta T + \sum_{t=1}^{\tau}\int_{E_t^0}^{E_t^1}\frac{V_{E_t}^H(.)}{V_y^H(.)}dE_t, \tag{8}$$

where a superscript H refers to a Hicksian or income-compensated function, $\Delta\pi$ and $\Delta\pi^A$ denote the changes in present value profit incomes, ΔT denotes the present value change in tax income, $V_{E_t}^H$ denotes the time t Hicksian marginal disutility of emissions, and V_y^H denotes the Hicksian marginal utility of present value income. The first term on the right-hand side is the sum of areas to the left of the Hicksian demand curves between initial and final consumer prices of electricity in the different periods, that is, the present value WTP (WTA) in exchange for the price decreases (increases). The second term expresses the sum of changes in present value consumer surpluses of the composite commodity, assuming that there is only a marginal change in its period t producer price dp_t^A for all t. The third

[19] In 2017, Sweden's installed capacity of wind power corresponded to around 1% of global capacity; https://www.natverketforvindbruk.se/sv/Fakta/.

[20] In December 2017, the Swedish government commissioned its Energy Agency to design such a stop mechanism; refer to Swedish Energy Agency (2018a).

term yields the present value change in profits of power plants as the green certificate (and possibly the electricity spot) price change. The fourth term yields the change in present value profit income from the composite sector. The fifth term represents the present value change in government tax revenue. The final term yields the present value WTP/WTA for any changes in pollution over the considered time frame. The ratio $V^H_{E_t}/V^H_y$ equals the partial derivative of the expenditure function with respect to E_t.

Eq. (8) is a line integral. This means that each WTP/WTA in the equation is calculated *conditional* on the previous values in order to incorporate the budget constraint, and the integral is path-independent in the sense that the order in which the terms are evaluated does not affect the magnitude of *CV*. Moreover, $CV \cdot \bar{V}_y = \Delta V$, where $\bar{V}_y > 0$ is the marginal utility of income evaluated at some present value income between the initial and final income levels, that is, *CV* is a sign-preserving measure of the *unobservable* utility change. Refer to Johansson and Kriström (2016, Ch. 6) for details on these matters.

Small or marginal price adjustments caused by the considered policy (if reasonably small) can be ignored in a general equilibrium analysis. To see why, consider a marginal change in the time t producer price p^A_t of the composite commodity:

$$\frac{\partial V(.)}{\partial p^A_t} dp^A_t = -V_y(.)\cdot\left[e^A_t(.)\cdot(1+t^m) - e^{As}_t(.) - e^A_t(.)\cdot t^m\right] dp^A_t$$
$$= V_y(.)\cdot[e^{As}(.) - e^A(.)]dp^A_t = 0 \qquad \forall t, \tag{9}$$

where e^{As}_t denotes the supply of the composite commodity, supply equals demand if the price clears the market, and the change in p^A_t is driven by the change in the policy parameter.

In the main text, drawing on Eq. (9), $\Delta\pi^A$ is cancelled from Eq. (4); the policy measure under evaluation is reasonably too small to have a significant impact on prices in the rest of the economy (and the policy change is so small that $e^A_t(.) \approx e^{AH}_t(.)$ for all t). Therefore, the terms $e^A_t(.)\cdot t^m dp^A_t$, $\forall t$, should be deducted from ΔT in Eq. (4), but in order to avoid unnecessary notational clutter, we assume that the terms are so small that they can be safely ignored. The same assumption applies to emissions, that is, $(\partial E_t/\partial p^A_t)dp^A_t \approx 0$ for all t. Finally, in Eq. (4), it is assumed that the WTP/WTA for changes in emissions can be approximated by point estimates based on Marshallian functions.

If the spot price of electricity adjusts marginally due to the change in the quota, one can proceed as in Eq. (9) and eliminate $\Delta\pi^{NG}$ from the *CV*-expression.

A.2. Empirics

In Section 5.4, reference is made to the WTP-study by Ma et al. (2015). They only report a perpetual WTP (based on an unreported point estimate), using a discount rate of 5%. The present value mean equal to USD 0.015 per kWh was converted to a 2018 SEK point estimate by multiplying by 0.05, adjusting for inflation (+13% since 2006), and assuming that USD/SEK = 8.84. This gives an estimate of 0.0088 SEK per kWh.

The environmental cost reported in Section 5.4 was estimated as follows. On average, a coal-fired power plant is considered to be marginal in the Nordic (Nord Pool) market. If the extra electricity in 2018, that is, Δe_{2018}, is generated by a Swedish coal-fired plant, Ecosense LE estimates that the cost to human health is about SEK 0.04 and about SEK 0.07 per kWh (if 1 EUR = 10.62), if damage to the natural environment is added. These estimates are considerably higher than the mean WTP reported by Ma et al. (2015). The estimate is based on Ecosense LE's default values for emissions caused by a hard coal (IGCC) 450 MW plant of NO_x, SO_2, PM_{10}, $PM_{2.5}$, NMVOC, and NH_3. Note that the plant is covered by the EU ETS, that is, there are no additional emissions of CO_2. We cannot rule out that a Danish coal-fired plant is marginal on the market. Nevertheless, Swedes may be altruists in the sense that they care for damage caused abroad. Loureiro and Loomis (2013) provide some empirical evidence that altruistic values (passive-use values) associated with, for example, environmental damage may go well beyond the territorial limits of affected countries. Refer also to the discussion of cross-border externalities in Johansson and Kriström (2016, pp. 26–28).

The range of environmental damages (indirectly) incurred by removing the certificates is then estimated to be in the interval (0.0088, 0.07) per kWh generated by a coal-fired plant in the scenarios. To account for the fact that e.g. wind power may generate negative externalities, or that Swedes ignore any damage they cause abroad if they import electricity, we set the lower limit to zero in the simulations.

A.3. Data used for the main results

The data used for the 2003–2017 period are displayed in Table 2.

Table 2
Data for 2003–2017, in 2017 prices (rounded). p^s_t = electricity spot price (Nord-Pool system price) at year t, p^d_t = approximated variable transmission tariff, t^e = unit electricity tax, p^c_t = certificate price, CPI = consumer price index (2017 = 1), and α_t denotes the quota. The value-added tax has been set to 0.25 throughout. Sources. Swedish Energy Agency and Statistical Bureau of Sweden. Units for p^s (electricity price), p^d, t^e and p^c are SEK per kWh. Annual quotas (α_t) are given as fractions.

	Year	p^s_t	p^d_t	t^e	p^c_t	CPI	α_t
1	2003	0.39	0.27	0.26	0.25	0.86	0.07
2	2004	0.30	0.28	0.28	0.25	0.87	0.08
3	2005	0.31	0.28	0.29	0.23	0.87	0.10
4	2006	0.51	0.28	0.30	0.19	0.88	0.13
5	2007	0.29	0.28	0.29	0.23	0.90	0.15
6	2008	0.46	0.27	0.29	0.35	0.93	0.16
7	2009	0.40	0.30	0.30	0.34	0.93	0.17
8	2010	0.54	0.31	0.30	0.27	0.94	0.18
9	2011	0.44	0.32	0.29	0.19	0.97	0.18
10	2012	0.28	0.33	0.30	0.17	0.98	0.18
11	2013	0.34	0.36	0.30	0.20	0.98	0.14
12	2014	0.28	0.37	0.30	0.18	0.97	0.14
13	2015	0.20	0.38	0.30	0.16	0.97	0.14
14	2016	0.26	0.39	0.30	0.14	0.98	0.23
15	2017	0.28	0.40	0.29	0.07	1	0.25

The assumptions used in the ex ante analysis are shown in Table 3. For convenient reference, we have added the ex post values (average values for the period 2003–2017).

Table 3
Parameter assumptions used in the Ex post and Ex ante analyses. p^s = mean electricity price in 2017 prices, t^e = mean energy tax 2003–2017, p^c = mean certificate price 2003–2017. Externality = externality cost SEK per kWh.

	Ex post	Ex ante
p^s	0.35	0.30
t^e	0.29	0.32
p^c	0.21	0.07
α	0.15	0.21
Externality	0.07	0.07

A.4. Parameter assumptions used for the simulations

For the certificate and electricity price simulations, we use the observed data for the period 2003–2017 to estimate a best fitting distribution. We tested a number of different distributions, using Von Mises, Anderson and Kolmogorov-Smirnov (KS) tests for distances between distributions. We also used goodness of fit statistics (Aikaike (AIC) and the Bayesian version (BIC)) The sample sizes are small and the robustness of these tests are known to be sensitive to sample size, not the least the KS test. The tests suggested the use of the log-normal for the electricity prices and the gamma distribution for the certificates. The data underlying our choices are displayed in Table 4.

Table 4
Statistical tests for fitting distributions.

	Weibull	Gamma	Lognorm	Normal
(a) Distribution tests for electricity prices 2003–2017. Unit = SEK/100 2017 SEK				
Von Mises	0.12	0.11	0.10	0.13
Andersson-Darling	0.68	0.59	0.55	0.71
Kolmogorov-Smirnov	0.22	0.21	0.20	0.23
Aikake	−26.55	−28.38	−28.75	−26.99
BIC	−25.14	−26.97	−27.33	−25.58
(b) Distribution tests for certificate prices 2003–2017. Unit = SEK/100 2017 SEK				
Von Mises	0.08	0.07	0.09	0.07
Andersson-Darling	0.51	0.48	0.60	0.47
Kolmogorov-Smirnov	0.20	0.17	0.17	0.20
Aikake	−26.55	−28.38	−28.75	−26.99
BIC	−25.14	−26.97	−27.33	−25.58

For the environmental externality, we use a uniform distribution and, as justified above, use the interval (0, 0.07) SEK. We set 1 EUR = 10.62 SEK. Finally, Fig. 4 displays the histograms of the price data together with the fitted parametric distributions.

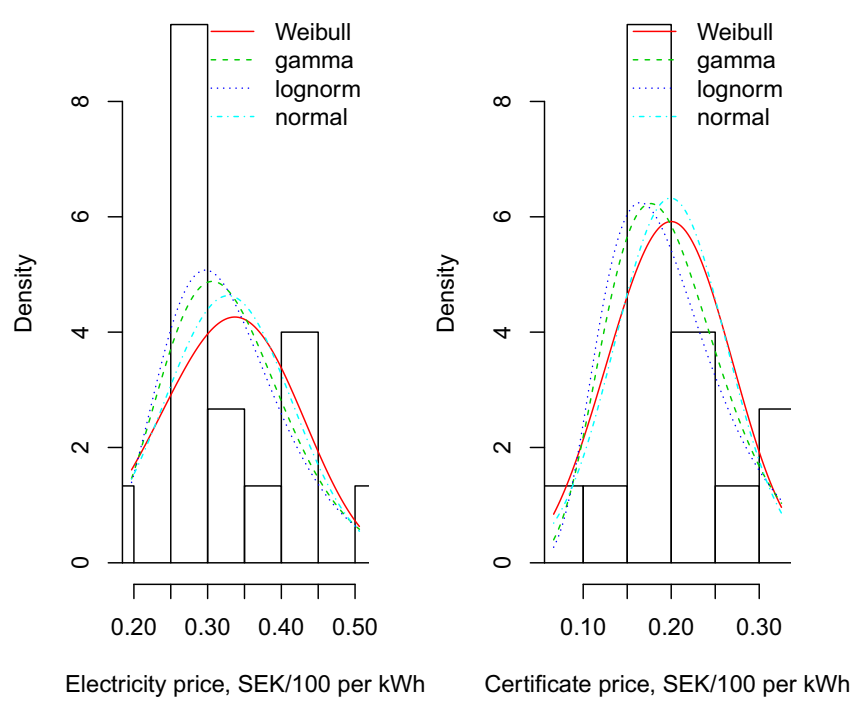

Fig. 4. Fitted densities for the prices used in simulations.

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